Testing Inheritance Patterns

Biology 2e chapter 12 reading connection | Teacher guide / classroom preparation

Learning target

How closely do observed kernel traits match an inheritance prediction?

Timing and materials

One class period; dry ears are reusable.

Documented monohybrid genetic corn ears, magnifier if useful, counting sheet, removable position marker and calculator. Do not substitute decorative corn with unknown parents.

Classroom route

1. Obtain ears from a documented cross and identify the trait and parent genotypes. Predict the phenotype ratio before counting.

2. Choose a systematic region or count the entire ear without removing kernels. Mark the starting position to avoid double counting. Count at least three ears and retain each ear's counts.

3. Pool counts only for comparable crosses. Compute expected counts from the proposed ratio, then calculate chi-square if appropriate; evaluate assumptions and sample size.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

For the defined Pp x Pp model, expected phenotypes are 3 purple:1 yellow. Expected counts equal total x 0.75 and total x 0.25. Chi-square summarizes squared deviations divided by expected counts; do not infer a p-value without the distribution and its assumptions.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

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Model limitations

Generated kernel populations and original artwork, not photographs or supplier data. Real corn pigment genetics can involve more than one locus; apply the prediction only to a documented suitable cross.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Genetic corn classroom materials

https://www.carolina.com/plant-genetics/monohybrid-genetics-with-corn-kit/176362.pr

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.